

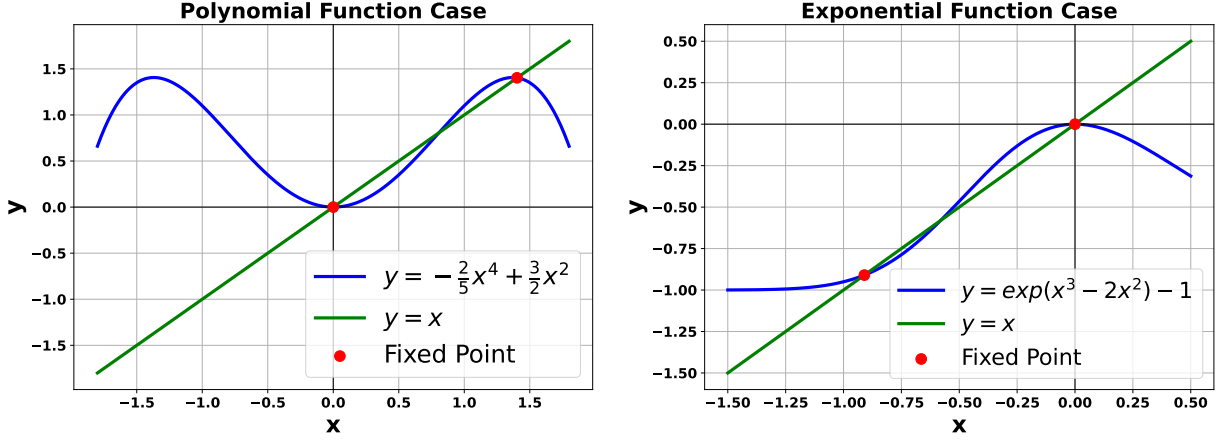


Figure 1: Example of polynomial (left) and exponential (right) functions contain at least two fixed points (red dots), and points near these fixed points will converge to them under fixed-point iteration.

Definition 3.1 (Fixed point, [AH09]). *Let D be a subset of $\mathbb{R}^d$. We say a function $f : D \rightarrow \mathbb{R}^d$ has a fixed point $p \in D$ if $f(p) = p$.*

Then, we introduce contractive mapping, which is a key concept in fixed point iteration convergence.

Definition 3.2 (Contractive mapping, Definition 5.1.2 of [AH09]). *Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$. Let D be a subset of $\mathbb{R}^d$. We say that a function $f : D \rightarrow \mathbb{R}^d$ is contractive with contractivity constant $K \in [0, 1)$ if*

$$\|f(x) - f(x')\| \leq K\|x - x'\|, \forall x, x' \in V.$$

Contractive mapping means that the function output space becomes ‘smaller’ than its input space. Then, we are able to introduce the Banach fixed-point theorem, the key tool in this work.

Lemma 3.3 (Banach fixed-point theorem, Theorem 5.1.3 of [AH09]). *Let $\|\cdot\|$ be a norm on $\mathbb{R}^d$. Let D be a nonempty closed set of $\mathbb{R}^d$. Suppose that $f : D \rightarrow \mathbb{R}^d$ is a mapping that satisfies the following*

- $f(x) \in D$ whenever $x \in D$.
- f is contractive with contractivity constant $K \in [0, 1)$.

Then it holds that

- The function f has a unique fixed point $p \in D$.
- For any initial point $x^{(0)} \in D$, the fixed-point iteration $x^{(t)} = f(x^{(t-1)})$, $t \geq 1$, converges to the fixed point p as $t \rightarrow \infty$.
- The following error bounds hold:

$$\|x^{(t)} - p\| \leq \frac{K^t}{1 - K} \|x^{(1)} - x^{(0)}\|,$$

$$\begin{aligned}\|x^{(t)} - p\| &\leq \frac{K}{1-K} \|x^{(t)} - x^{(t-1)}\|, \\ \|x^{(t)} - p\| &\leq K \|x^{(t-1)} - p\|.\end{aligned}$$

Lemma 3.3 told us the sufficient condition for a single unique fixed point. Although checking for contractivity is usually difficult, for differentiable functions, it becomes easier by examining the Jacobian matrix. The following lemma is a corollary of Lemma 3.3.

Lemma 3.4 (Banach fixed point theorem, vector case, informal version of Lemma A.5). *Let $D \subseteq \mathbb{R}^d$ be a nonempty closed set. Suppose that $f : D \rightarrow \mathbb{R}^d$ is differentiable and satisfies the following:*

- $f(x) \in D$ whenever $x \in D$.
- There exists constant $K < 1$ such that for every $i \in [d]$,

$$\left\| \frac{df_i(x)}{dx} \right\|_1 \leq K, \quad \forall x \in D$$

where $f_i(x)$ is the i -th entry of $f(x)$.

Then it holds that

- The function f has a unique fixed point $p \in D$.
- For any initial point $x^{(0)} \in D$, the fixed point iteration $x^{(t)} = f(x^{(t-1)})$, $t \geq 1$, converges to the fixed point p as $t \rightarrow \infty$.
- The following error bounds hold:

$$\begin{aligned}\|x^{(t)} - p\|_\infty &\leq \frac{K^t}{1-K} \|x^{(1)} - x^{(0)}\|_\infty, \\ \|x^{(t)} - p\|_\infty &\leq \frac{K}{1-K} \|x^{(t)} - x^{(t-1)}\|_\infty, \\ \|x^{(t)} - p\|_\infty &\leq K \|x^{(t-1)} - p\|_\infty.\end{aligned}$$

When $d = 1$, Lemma 3.4 boils down to the scalar case.

3.3 Looped Neural Networks

In this section, we give the formal definition of Looped Neural Network.

Definition 3.5 (Looped Neural Network). *Let $W = [w_1, \dots, w_d]^\top \in \mathbb{R}^{d \times d}$ be a weight matrix and $b \in \mathbb{R}^d$ be the bias parameter. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable activation function. We consider the one layer of neural network in the form*

$$f(x; W, b) := g(Wx + b)$$

where g is applied entry-wise. The L -layer looped neural network is defined as

$$\begin{aligned}\text{NN}(x^{(0)}; W, b, L) &:= x^{(L)} \\ x^{(t)} &:= f(x^{(t-1)}; W, b), \quad \forall t \in [L].\end{aligned}$$

Remark 3.6. *Note that the Looped Neural Network (LNN) shares similarities with Recurrent Neural Networks (RNN) but is slightly different. RNN maps sequence to sequence, which means in each loop, RNN has new input data, while LNN does not. Furthermore, RNN generates output in each loop while LNN only outputs after all loops.*

4 MAIN RESULTS

We first introduce our general theorem for looped neural networks in Section 4.1 and then introduce our robust version in Section 4.2.

4.1 General Theorem

We have the following general theorem which provides a sufficient condition for the existence of multiple fixed points for Looped Neural Network.

Theorem 4.1 (General result). *Consider the L -layer looped neural network $\text{NN}(x^{(0)}; W, b, L)$ defined in Definition 3.5. If the following conditions hold:*

- *There exists disjoint $D_1, \dots, D_m \subseteq \mathbb{R}^d$ such that for every $i \in [m]$, and for every $x \in D_i$, $\text{NN}(x; W, b, 1) \in D_i$.*
- *The weight matrix W and the activation function g satisfy: For every $i \in [m]$, there exists $K_i \in [0, 1)$ such that for every $x \in D_i$, and for every $j \in [d]$,*

$$|g'(\langle w_j, x \rangle)| \cdot \|w_j\|_1 \leq K_i.$$

Then, the following statements hold:

- *The single layer of the looped neural network, $f(x; W)$, has at least m fixed-points $p_1, \dots, p_m$ satisfying : For every $i \in [m]$, there exists a constant $\epsilon_i > 0$, for any initial point $x^{(0)} \in D_i$ with $\|x^{(0)} - p_i\|_\infty \leq \epsilon_i$, we have*

$$\lim_{L \rightarrow \infty} \text{NN}(x^{(0)}; W, b, L) = p_i.$$

- *For every $i \in [m]$, there exists a constant $c_i > 0$ such that for any $L \geq 2$,*

$$\|\text{NN}(x^{(0)}; W, b, L) - p_i\|_\infty \leq K_i^L \cdot c_i \epsilon_i.$$

Proof. Assume the conditions in the statement hold. Fix $i \in [m]$. Then we have $f(x; W) = \text{NN}(x; W, b, 1) \in D_i$ for every $x \in D_i$. Next, for every $j \in [d]$, and for the j -th entry of $f(x; W)$, where we denoted by $f_j(x; W)$, we have

$$\begin{aligned} \left\| \frac{df_j(x; W, b)}{dx} \right\|_1 &= \left\| \frac{dg(\langle w_j, x \rangle)}{dx} \right\|_1 \\ &= \|g'(\langle w_j, x \rangle) \cdot w_j\|_1 \\ &= |g'(\langle w_j, x \rangle)| \cdot \|w_j\|_1 \\ &\leq K_i, \end{aligned}$$

where the first step follows from the definition of $f_j(x; W)$, the second step uses the chain rule, the third step is due to the property of norm, and the last step uses the condition in lemma.

Therefore, we can apply Lemma 3.3, for each $i \in [m]$, there exists a fixed point p_i which can be converged to by the fixed point iteration, and thus it can be found by the looped neural network with initial point $x^{(0)}$ when the number of layers goes to infinity. Next, for every $i \in [m]$, and for an initial point $x^{(0)} \in D_i$, there exists $\epsilon_i := \sup_{y, z \in D_i} \|y - z\|_\infty$, $c_i := \frac{1}{1-K_i}$ such that we have

$$\|\text{NN}(x^{(0)}; W, b, L) - p_i\|_\infty = \|x^{(L)} - p_i\|_\infty$$

$$\begin{aligned}
&\leq \frac{K_i^t}{1 - K_i} \|x^{(1)} - x^{(0)}\| \\
&\leq K_i^t \cdot c_i \epsilon_i,
\end{aligned}$$

where the first step follows Definition 3.5, and the second step uses the error bound in Lemma 3.3, and the second steps follows the definition of ϵ_i and c_i . $\square$

Theorem 4.1 gives us a way how to find different fixed points of vector-valued functions such as looped neural networks. Note that for different inputs, the fix point iteration behavior may be different, even when the model weights are fixed. We refer readers to Figure 1 and Figure 2 for more intuition.

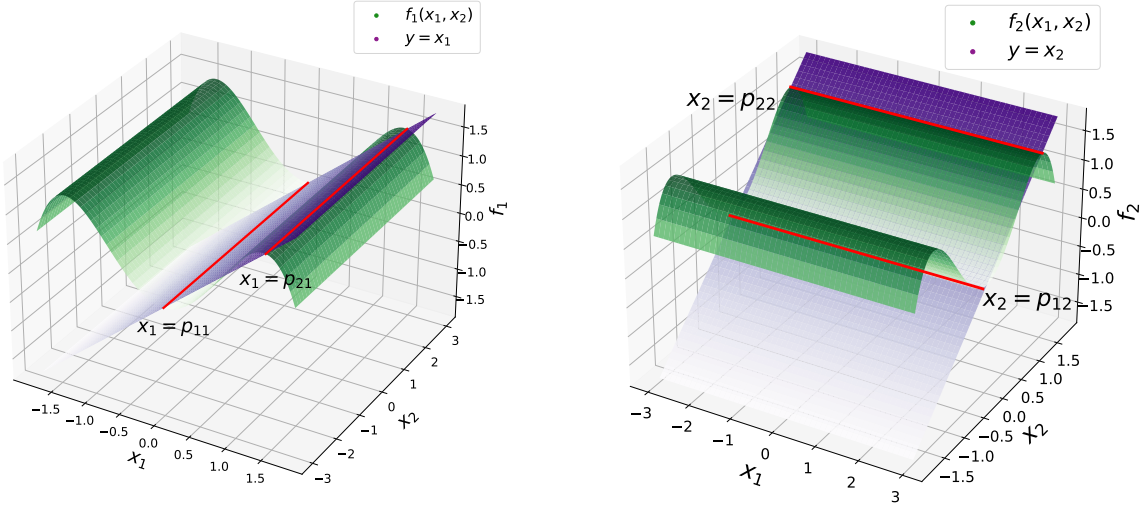


Figure 2: Example of a 2-d case of Theorem 5.1. **Left:** The graph of $f_1(x_1, x_2)$ (in green) and $y = x_1$ (in purple), with the red line indicating the intersection of two curves (1-th dimension fixed point). **Right:** The graph of $f_2(x_1, x_2)$ (in green) and $y = x_2$ (in purple), with the red line indicating the intersection of two curves (2-th dimension fixed point).

4.2 Perturbed Fixed Point Iteration

Next, we consider a variant of fixed point method, where each iteration there is noise term $h(x)$.

Theorem 4.2 (Robust Banach fixed point theorem, scalar case). *Let $D \subseteq \mathbb{R}$ be a nonempty closed set. Suppose that $f : D \rightarrow \mathbb{R}$ is differentiable and satisfies the following:*

- $f(x) \in D$ whenever $x \in D$.
- There exists constant $K \in [0, 0.95]$ such that

$$|f'(x)| \leq K, \quad \forall x \in D.$$

- For any initial point $x^{(0)} \in D$, consider the perturbed fixed point iteration $x^{(t)} = f(x^{(t-1)}) + h(x^{(t-1)})$, for any $t \geq 1$, where the function h satisfies $|h(x)| \leq 1/m$ for every $x \in D$ for some sufficiently large $m > 0$.

Then it holds that

- The function f has a unique fixed point $p \in D$.
- The following error bounds hold:

$$\begin{aligned} |x^{(t)} - p| &\leq K|x^{(t-1)} - p| + \frac{1}{m}, \\ |x^{(t)} - p| &\leq K^t|x^{(0)} - p| + \frac{20}{m}. \end{aligned}$$

Proof. Clearly f has a fixed point $p \in D$ by Lemma A.4.

We first show the first error bound. We can show that

$$\begin{aligned} |x^{(t)} - p| &= |f(x^{(t-1)}) - p + h(x^{(t-1)})| \\ &= |f(x^{(t-1)}) - f(p) + h(x^{(t-1)})| \\ &= |f'(c) \cdot (x^{(t-1)} - p) + h(x^{(t-1)})| \\ &\leq |f'(c) \cdot (x^{(t-1)} - p)| + |h(x^{(t-1)})| \\ &= |f'(c)| \cdot |x^{(t-1)} - p| + |h(x^{(t-1)})| \\ &\leq K|x^{(t-1)} - p| + \frac{1}{m} \end{aligned}$$

where the first step uses the perturbed fixed point iteration $x^{(t)} = f(x^{(t-1)}) + h(x^{(t-1)})$, the second step is due to the fact that p is a fixed point of f , the third step follows from the mean value theorem where c is a point between $x^{(t-1)}$ and p , the fourth step uses the triangle inequality, the fifth step follows from basic algebra, and the last step follows from $|f'(x)| \leq K$ and $|h(x)| \leq 1/m$.

Next, we show the second error bound. We can show that

$$\begin{aligned} |x^{(t)} - p| &\leq K|x^{(t-1)} - p| + \frac{1}{m} \\ &\leq K(K|x^{(t-2)} - p| + \frac{1}{m}) + \frac{1}{m} \\ &\leq \dots \\ &\leq K^t|x^{(0)} - p| + \sum_{i=1}^{t-1} \frac{K^i}{m} \\ &\leq K^t|x^{(0)} - p| + \frac{1 - K^t}{(1 - K)m} \\ &\leq K^t|x^{(0)} - p| + \frac{20}{m}, \end{aligned}$$

where the first four steps follow from recursively using the third error bound, the fifth step is due to the sum of geometric series, and the last step follows from $K \in [0, 0.95]$. $\square$

Theorem 4.2 shows that each fix iteration process is robust and may not be hurt by the noise much. It corresponds to the setting that deep neural networks with residual connections [HZRS16], where after each layer the hidden states are only slightly perturbed. Our experiments in Section 6 support our theoretical analysis of robustness.